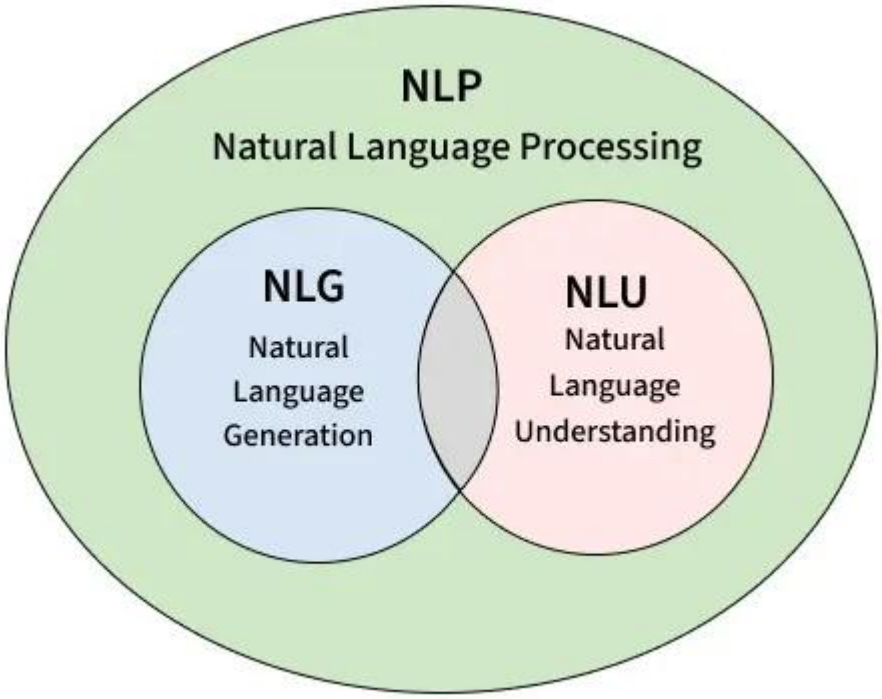
INTRODUCTION NATURAL LANGUAGE GENERATION:

Natural Language Generation (NLG) is a subfield of **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)** that enables computers to automatically generate meaningful, human-readable text from structured or unstructured data.



NLG generates human-like text from structured data such as tables, knowledge graphs, or events. It is the **reverse of NLU**: NLU converts human language into machine-understandable information, while NLG converts machine information into **human-readable text**.

How NLG Works

The main stages of an NLG system are:

1. **Content Determination** – Selects the important information that should be included.
2. **Document Structuring** – Arranges the selected information in a logical order.
3. **Aggregation** – Combines related information to avoid repetition.
4. **Lexicalization** – Selects suitable words and expressions.
5. **Referring Expression Generation** – Generates references such as *he, she, it, they*, or names.
6. **Linguistic Realization** – Applies grammar rules to form correct and natural sentences.

NLG vs NLU vs NLP

Aspect	NLP	NLG	NLU
Meaning	Processes human language	Generates human language	Understands human language
Input	Text or speech	Structured data/information	Natural language
Output	Processed language/information	Human-readable text	Machine-readable meaning
Main Goal	Process language	Produce language	Understand meaning
Examples	Translation, summarization	Reports, chatbot responses	Intent detection, sentiment analysis

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Architecture of NLG Systems:

Natural Language Generation (NLG) architecture describes how a system converts structured data or information into meaningful natural-language text.

Architecture of NLG Systems:

Input Data

↓

1. Content Determination

↓

2. Document Structuring

↓

3. Aggregation

↓

4. Lexicalization

↓

5. Referring Expression

↓

6. Linguistic Realization

↓

Natural Language Output

Components of NLG Architecture:

1. Content Determination

The system selects **important information** from the input data.

Example: From weather data, it selects temperature, rainfall, and wind information.

2. Document Structuring

The selected information is **organized in a logical order**.

Example: Temperature → rainfall → weather condition.

3. Aggregation

Related information is **combined into a single sentence** to avoid repetition.

Example:
Instead of: *"Temperature is 30°C. The temperature is high."*
It generates: *"The temperature is high at 30°C."*

4. Lexicalization

The system chooses **appropriate words and phrases**.

Example: It may choose *"heavy rain"* instead of *"a large amount of rainfall."*

5. Referring Expression Generation

The system decides how to refer to people, objects, or entities using **names, pronouns, or descriptions**.

Example: *"Ravi submitted the report. He received a confirmation."*

6. Linguistic Realization

The system applies **grammar, sentence structure, tense, and punctuation** to produce a correct sentence.

Example:
Input: Temperature = 35°C
Output: **"The temperature is 35°C today."**

Input Data → Select Information → Organize → Combine → Choose Words → Apply Grammar → Natural Language Output

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Generation Tasks and Representations in NLP:

Generation in NLP means producing **natural-language text from some input or internal representation**. The input may be data, meaning, a sentence representation, or information extracted from a database.

1. Generation Tasks in NLP:

Generation tasks are the different activities where an NLP system produces text.

1. Text Generation
The system generates new text based on a given input or topic.
Example:
Input: Topic = Artificial Intelligence
Output:
“Artificial Intelligence enables computers to perform tasks that normally require human intelligence.”
Applications:

- Content creation
- Story generation
- Question answering
- Chatbots

2. Machine Translation
Machine translation converts text from **one language to another**.
Example:
English: "Good morning"
↓
NLP System
↓
Tamil: "காலை வணக்கம்"
Applications:

- Google Translate
- Multilingual websites
- Language-learning applications

3. Text Summarization
The system creates a **short version of a long document** while keeping the important information.
Example:
Original:

“Artificial Intelligence is a technology that allows machines to learn, reason, and perform tasks that usually require human intelligence.”
Summary:
“AI enables machines to perform human-like intelligent tasks.”
There are two common types:

- **Extractive summarization** – selects important sentences from the original text.
- **Abstractive summarization** – generates new sentences based on the meaning.

4. Dialogue Generation
The system generates responses during a **conversation with a user**.
Example:
User: What is NLP?
↓
System: NLP is a field of AI that helps computers understand and process human language.
Applications:

- Chatbots
- Virtual assistants
- Customer support systems

5. Question Answering
The system generates an answer to a user's question.
Example:
Question: What is the capital of India?
↓
Answer: New Delhi.
The answer may be selected from existing information or generated using a language model.

6. Data-to-Text Generation
The system converts **structured data into natural-language text**.
Example:
Input:
Temperature = 32°C
Rain = 80%
Output:
“Today, the temperature is 32°C with an 80% chance of rain.”
Applications:

- Weather reports
- Financial reports
- Sports reports
- Business dashboards

7. Image-to-Text Generation
The system generates a description of an image using computer vision and NLP.
Example:
Image → AI System → "A dog is playing in the park."
Application:

- Image captions
- Photo search

2. Representations in NLP:

Before generating text, computers usually represent language in a **machine-readable form**.

A representation describes the **meaning, structure, or information** that the system needs to generate the final text.

Common NLP Representations:

1. Words and Tokens

Text is divided into smaller units called **tokens**.

Example:

“The student reads a book.”

Tokens:

The | student | reads | a | book

Tokens are basic units used by NLP systems.

2. Part-of-Speech Representation

Each word can be represented according to its grammatical category.

Example:

The → Determiner

student → Noun

reads → Verb

a → Determiner

book → Noun

Common POS categories include:

- Noun
- Verb

- Adjective
- Adverb
- Pronoun
- Preposition
- Determiner

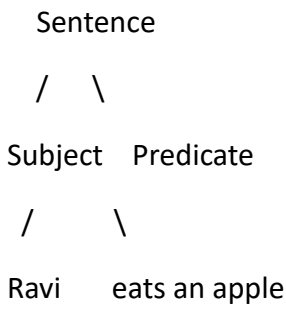
This helps the system understand sentence structure.

3. Syntax Representation

Syntax represents **how words are connected in a sentence**.

Example:

“Ravi eats an apple.”



Syntax helps the system generate grammatically correct sentences.

4. Semantic Representation

Semantic representation describes the **meaning of a sentence**.

Example:

“Ravi eats an apple.”

Can be represented as:

Person = Ravi

Action = Eat

Object = Apple

This tells the system **who performs the action and what the action affects**.

5. Logical Representation

Information can also be represented using logical statements.

Example:

“All students attend classes.”

Representation:

For every student → attends → class

Logical representation is useful for reasoning and knowledge-based systems.

6. Feature-Based Representation

A sentence or word can be represented using different features.

Example:

Word: boy

Features:

Gender = Male

Number = Singular

Person = Third

These features help in generating grammatically correct text.

7. Semantic Frames

A **semantic frame** represents an event using important roles.

Example:

“Ravi bought a laptop from the shop.”

Event: Buying

Buyer → Ravi

Object → Laptop

Source → Shop

This representation is useful for generating descriptions of events.

8. Knowledge Graph Representation

Information can be represented as **entities and relationships**.

Example:

Ravi — studies at — College

Ravi — lives in — Chennai

The system can use this information to generate:

“Ravi studies at a college and lives in Chennai.”

Application of NLG:

NLG (Natural Language Generation) is a branch of **NLP (Natural Language Processing)** that enables computers to automatically generate meaningful and understandable human-like language from data, information, or structured input.

Important Applications of NLG:

1.Chatbots:

NLG is used to generate natural responses in conversations with users.

- Provides answers to user questions.
- Generates conversational responses.
- Makes interactions more human-like.
- Used in customer support systems.

Examples: ChatGPT, Siri, and Alexa.

2.Automatic Report Generation

NLG can convert structured data into readable reports.

- Generates financial reports.
- Creates business performance summaries.
- Produces weather reports.
- Reduces the need for manual report writing.

Example:

Data → “Sales increased by 20%”

NLG → “The company’s sales increased by 20% this time”

3. Text Summarization

NLG is used to generate short and meaningful summaries from large amounts of text.

- Summarizes news articles.
- Creates research-paper summaries.
- Summarizes business documents.
- Helps users understand lengthy information quickly.

4. Machine Translation

NLG helps generate text in the target language after translating information from a source language.

Example:

English: “I am going to college.”
Tamil: “நான் கல்லூரிக்குச் செல்கிறேன்.”
NLG helps produce a natural and grammatically appropriate output.

5. Content Generation

NLG can automatically create different types of written content.

- Articles
- Blog posts
- Product descriptions
- Emails
- Social media posts
- Advertisements

This helps save time and effort in content creation.

6. Question Answering Systems

NLG generates natural-language answers to questions asked by users.

Example:

Question: “What is the capital of India?”
Generated answer: “The capital of India is New Delhi.”
It is commonly used in search systems, educational applications, and virtual assistants.

7. Healthcare Applications

NLG can convert medical data into understandable text.

- Generates Medical case summaries.
- Creates medical reports.

example :

Patient data → NLG → “The patient’s blood pressure is normal.”

8. Education

NLG is useful for generating personalized educational content.

- Creates questions and answers.
- Generates explanations.
- Provides feedback to students.
- Creates study materials.
- Supports intelligent tutoring systems.

Example:

A learning system can generate a simple explanation of a difficult topic based on a student's level.

9. Customer Service

NLG helps organizations automatically communicate with customers.

- Provides order-status updates.
- Generates notifications.
- Creates personalized messages.

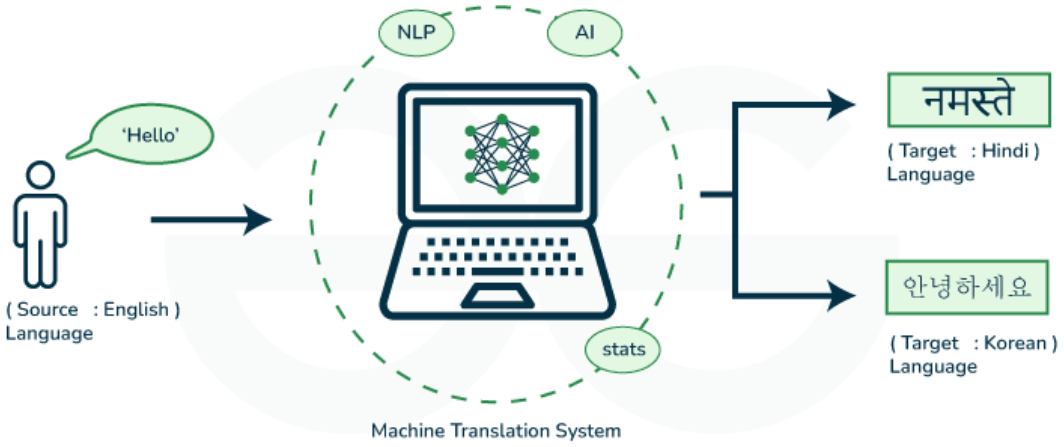
Example:
“Your order has been shipped and will arrive tomorrow.”

Machine Translation:

Machine Translation (MT) is the use of **AI and machine learning** to automatically translate text or speech from one language to another. It helps people communicate across different languages and is widely used in **translation apps, websites, chatbots, and multilingual communication**.

What is Machine Translation?

Machine Translation (MT) is a branch of **NLP** that uses computers to automatically translate **text or speech from one language to another** while storing the original meaning and grammar.



Machine Translation Model

History of Machine Translation:

Machine Translation (MT) means automatically translating text from a **source language** into a **target language** using computers. It was one of the First applications of computers.

Three Main Uses of Machine Translation

1. **Basic Translation**
    - Gives the general meaning of a foreign text.
    - It may contain some mistakes.
    - Example: Free online translation services.
2. **Human-Edited Translation**
    - The original text is written in a simple and limited language.
    - The computer translates it.
    - A person then checks and corrects the translation.
3. **Limited -Source Translation**
    - Translation is completely automatic.
    - It is used for simple and fixed types of language.
    - **Example:** Weather reports.

Key Approaches in Machine Translation

There are **four main approaches** to Machine Translation:

1. **Rule-Based Machine Translation (RBMT)**
    - Uses **grammar rules and dictionaries** to translate text.
    - Follows predefined related rules.
2. **Statistical Machine Translation (SMT)**
    - Uses **machine learning and statistical patterns** from previously translated texts.
    - Selects the most probable translation.
3. **Neural Machine Translation (NMT)**
    - Uses **neural networks** to translate text.
    - Learns meaning and patterns from large amounts of data.
    - Produces more natural translations.
4. **Hybrid Machine Translation**
    - Combines **two or more approaches**, usually rule-based and statistical or neural methods.
    - Aims to improve translation accuracy.

Why We Need Machine Translation in NLP

1. **Better Communication** – Helps people who speak different languages communicate easily.
2. **Increases Accessibility** – Makes online content available in different languages.
3. **Improves Efficiency** – Translates large amounts of text quickly.
4. **Helps Language Learning** – Helps learners understand new words and phrases.

Applications of Machine Translation

1. **Communication** – Helps people communicate across different languages.
2. **Localization** – Translates websites, software, and digital content.
3. **Business** – Translates documents, contracts, and business materials.
4. **Education** – Helps students learn new languages.
5. **Government** – Translates official documents and communications.

Problems in Machine Translation:

1. **Language Differences**
   There are many languages. Each language has different words, grammar, and culture. This makes NLP difficult.
2. **Training Data**
   NLP needs a large amount of good-quality data to train the model.
3. **Development Time and Resources**
   NLP projects may need more time, good data, expert people, and powerful computers.
4. **Phrasing Ambiguity**
   One sentence or phrase can have different meanings. NLP must use the context to understand the correct meaning.
5. **Misspellings and Grammar Errors**
   People may make spelling and grammar mistakes. NLP must find and correct these errors.
6. **Preference in NLP**
   Training data may contain unfair bias. NLP systems need diverse and fair data.
7. **Words with Multiple Meanings**
   One word can have different meanings in different situations. NLP must understand the correct meaning from the context.
8. **Multiple Languages**
   NLP should understand many languages. It needs multilingual data and translation methods.
9. **Possibility of Error and False Positives**
   Sometimes NLP gives a wrong answer or prediction. Confidence scores and proper thresholds can help reduce errors.
10. **Continuous Conversations**
    NLP systems should remember the conversation and understand the user's meaning while talking continuously.